Experiment 2, Part B: Plasmid DNA Purification (Miniprep and Midiprep)

Purpose

Purify plasmid DNA from the overnight cultures of Part A (binder plates, by miniprep) and from the target and reporter cultures of Part C (by midiprep) for use in the split-transcription-factor screen.

Materials and equipment

- Plasmid miniprep reagents and plates from a Zymo plasmid miniprep kit, used in a team-adapted 96-well format: buffers P1, P2 and P3, binding buffer, W1 and W2, any 96 well filter plate (blue) and any silica DNA-binding filter plate (red)
- ZymoPURE II Plasmid Midiprep Kit
- 96-well round-bottom deep-well plates, PCR plate and foil seals
- Plate centrifuge, plate shaker and spectrophotometer
- Nuclease-free water

Notes before starting

- Do not use a vacuum manifold. Wherever a step calls for vacuum, use the centrifuge instead.
- Do not use a silicone mat. Mix only by pipetting.
- “Evenly distribute” means splitting a volume between duplicate wells. For example, 250 µL of P1 is added as 125 µL to each of two duplicate plates, and the two are pooled later.

Procedure: miniprep

1. Grow 1 mL cultures in LB with carbenicillin (1:1,000 dilution of stock) in 96-well round-bottom deep-well plates for 12–16 hours at 37 °C and 1,000 rpm. If duplicate culture plates are grown, keep them separate until step 4. Make glycerol stocks of the cultures.
2. Harvest cells at 4,000 ×g for 5 minutes and discard the medium. Do not consolidate cultures at this stage.
3. Add 250 µL of P1 per index, distributed evenly between duplicate plates. Shake at 1,000 rpm for a few minutes until the pellets are fully resuspended.
4. If duplicate plates were used, consolidate the resuspended cells into one plate.
5. Add 250 µL of P2 per well and mix gently. Do not incubate for longer than 3 minutes.
6. Add 250 µL of P3 per well and mix gently. Make sure no purple colour remains.
7. Optional: spin the lysate at 4,000 ×g for 5–10 minutes to remove most cell debris.
8. Transfer everything to the filter plate (blue). If the spin was skipped, let the plate sit for about 5 minutes so the precipitate rises before spinning; filtering takes much longer otherwise.
9. Filter into a deep-well plate by centrifuging at 4,000 ×g for 5 minutes.
10. Add 190 µL of binding buffer per well and mix by pipetting.
11. Transfer to the silica DNA-binding filter plate (red). Centrifuge at 2,000 ×g for 1–2 minutes and discard the flow-through.
12. Add 800 µL of W1 per well. Centrifuge at 2,000 ×g for 1–2 minutes and discard the flow-through.
13. Add 800 µL of W2 per well. Centrifuge at 2,000 ×g for 1–2 minutes and discard the flow-through.
14. Add 200 µL of W2 per well. Centrifuge at 2,000 ×g for 1–2 minutes and discard the flow-through.
15. Centrifuge once more at 2,000 ×g for 1–2 minutes to remove excess wash buffer.
16. Place the silica plate on top of a PCR plate, using an inverted P1000 tip rack as a spacer.
17. Add 85 µL of water. Cover with foil and incubate at room temperature for 1–2 minutes.

18. Centrifuge at 4,000 ×g for 5 minutes and check that the expected volume was recovered.
19. Measure DNA concentration and purity (see Expected readout).

Procedure: midiprep

1. Grow overnight cultures of each target and the reporter in 50 mL of LB with carbenicillin (1:1,000 dilution of stock).
2. Purify plasmid DNA with the ZymoPURE II Plasmid Midiprep Kit, following the manufacturer's centrifuge protocol: https://au.zymoresearch.com/products/zymopure-ii-plasmid-midiprep-kit
3. Elute in 200 µL of nuclease-free water.
4. Measure DNA concentration and purity.

Controls

None.

Expected readout

DNA concentration and purity by spectrophotometer. Miniprep acceptance: A260/A280 of about 1.8 and A260/A230 of 2.0–2.2, and the concentration should be anything more than 80 ng/ul.